

Yu-Heng Lai

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Education

National Taiwan University — Taipei, Taiwan Aug. 2023 – Jun. 2027 (Expected)
B.S. in Electrical Engineering, Double Major in Physics GPA: 4.12/4.30

University of California, San Diego — San Diego, CA Sep. 2026 – Dec. 2026
Exchange Student, Electrical and Computer Engineering

Coursework: Machine Learning, Deep Learning, Computer Vision, Probability and Statistics, Linear Algebra

Research Experience

Spatiotemporal Machine Learning Lab, UC San Diego Sep. 2026 – Present
Undergraduate Researcher / Advisor: Prof. Rose Yu

- Designing a spatiotemporal forecasting benchmark to evaluate where, when, and how real-world events affect economic regions connected by a graph.
- Curated 22 economic and financial time-series datasets with contextual information; built a USDA grain-logistics graph with 19 markets and 36 state-to-port edges.

MAGICS Lab, Northwestern University (Remote) Oct. 2025 – Present
Undergraduate Researcher / Advisor: Prof. Han Liu

- Investigating the optimal balance between synthetic and real training data through mathematical modeling.
- Developing a PyTorch generative model to predict single-cell gene-expression responses to genetic perturbations.

Wireless Mobile Network Lab, National Taiwan University Feb. 2025 – Present
Undergraduate Researcher / Advisor: Prof. Hung-Yu Wei

- Analyzing experimental wireless measurements using time-series methods to inform model design.
- Developing deep generative models to synthesize wireless time-series data for digital-twin development.

Vision and Learning Lab, National Taiwan University Sep. 2025 – Feb. 2026
Undergraduate Researcher / Advisor: Prof. Yu-Chiang Wang

- Surveyed recent advances in video diffusion-based world models.
- Analyzed how world-model architectures support autoregressive rollout and spatial memory.

Selected Projects

EEG-Based Dementia Classification | Python, PyTorch, scikit-learn Sep. 2025 – Jan. 2026

- Built a two-stage SVM classifier for EEG from 88 subjects; achieved **74% accuracy** across control, Alzheimer's disease, and frontotemporal dementia on a 27-subject test set.
- Extracted graph signal processing features and compared classical classifiers with a CNN-Transformer; collaborated with a resident physician to interpret results.

Semiconductor Intelligence Platform | Python, Vertex AI Feb. 2026
3rd Place, 2026 TSMC IT CareerHack, Group B

- Built LLM pipelines to summarize 300+ news articles and 100 earnings-call transcripts for a 20-company intelligence platform; integrated a Vertex AI chatbot for company-specific Q&A.

Technical Skills

- **Languages:** Python, C/C++, MATLAB, TypeScript, JavaScript, HTML/CSS
- **ML & Scientific Computing:** PyTorch, scikit-learn, pandas, NumPy, SciPy
- **Signal & Graph Processing:** MNE, PyGSP, NetworkX
- **Visualization:** Matplotlib, seaborn
- **Tools & Platforms:** Git, Jupyter, Google Cloud Platform (Vertex AI), VS Code